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UNIVERSITY OF CAPE COAST
COLLEGE OF AGRICULTURE AND NATURAL SCIENCES
SCHOOL OF PHYSICAL SCIENCES
DEPARTMENT OF PHYSICS*
FIRST SEMESTER: 2022/2023
PHY 101: GENERAL PHYSICS I (THEORY)– GROUP 15
QUIZ 2 - DURATION: =40 MINS
TOTAL (20 MARKS) - ANSWER ALL QUESTIONS
INDEX NUMBER.:……. NAME:…….
DO THE ROUGH WORK OR WORKINGS THE QUESTION PAPER.

Use for Q1-Q3.

A 5.0g bullet moving at 100m/s strikes a log. Assume that the bullet undergoes a uniform deceleration and stops at 6cm.

Q1. What the time does it take for the bullet to stop in the log? **Ans ……**

Q2. What is the impulse on the log? Ans ……

Q3. What is the average force experienced by the log? Ans ……

Use for Q4.

A ball of 0.4kg mass and a speed of 3m/s has a head-on, completely elastic collision with a 0.6kg mass initially at rest.

Q4. Find the speeds on both bodies after the collision.

$v_1 = \ldots$

and

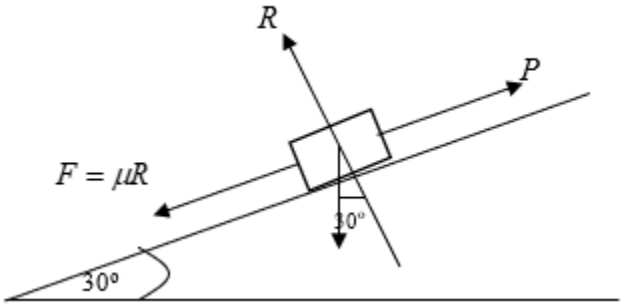
$v_2 = \ldots$

Use for Q5.

A body of mass 8kg is on the point of slipping down a plane of inclination 30°, what force applied parallel to the plane will just move it up the plane? [Take g=10ms^{-2}]

Q5. Ans….

Use for Q6.



Q6. What is $\mu = \ldots.$

Q7. Mr. Tetteh and his driver are traveling to Cape Coast in a Toyota pickup which is moving at 90 km/h down a straight highway. The driver of the pickup sees an accident in the distance and slows down to 40 km/h in 5.0 s. What is the average acceleration of the Toyota pickup?

Ans ……

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Q8. If a rabbit's velocity is $v = -0.62t^2\mathbf{i} + 7.2\mathbf{j}$, find the acceleration $\mathbf{a}$ at the time $t = 5s$, in unit-vector notation and in magnitude notation.
Ans ……
Ans ……
Q9. A car traveling at a speed of +12 m/s accelerates at a rate of +4 m/s^2 for a time of 5s. What is the car's speed at the end of 5s? **Ans ……**
Use for Q11-Q14.
A ball is dropped from rest from a high window of a tall building and falls to the ground in 3s. Neglecting air resistance. [Assume that g = 10 m/s].
Q11. What is the speed of the ball just before it hits the ground? **Ans ……**
Q12. What is the distance the ball has fallen from where it was released till it hit the ground?
Ans ……
Q13. What is the ball's acceleration? **Ans ……**
Q14. What distance will it travel in the last second (i.e. between t = 2s and t = 3s)? **Ans ……**
Use for Q15-Q17.
A 200 g mass vibrates horizontally without friction at the end of a horizontal spring for which k = 7.0 N/m. The mass is displaced 5.0 cm from equilibrium and released
Q15. What the maximum speed of the body? **Ans ……**
Q16. What is the speed of the body when it is 3.0 cm from the equilibrium position? **Ans ……**
Q18. A ball of mass 1.5kg is thrown vertically upwards from the ground with an initial velocity of 100ms^{-1}. Calculate the potential energy at a point A, 50m above the ground. **Ans ……**
Q19. Find the work done in moving an object along a straight line from (3, 2, -1) to (2, -1, 4) in a force field given by F = 4i – 3j +2k. Calculate the power dissipated if it took 4s to move the object to its final position. **Ans ……**
Ans ……
Q20. How much work is done by an applied force to lift a 15N block 3.0m vertically at a constant speed?
Ans ……